

EXPERIMENT 23: Write a Procedure To Attach Virtual Block To The Virtual Machine And Check Whether It Holds the Data Even After The Release Of The Virtual Machine

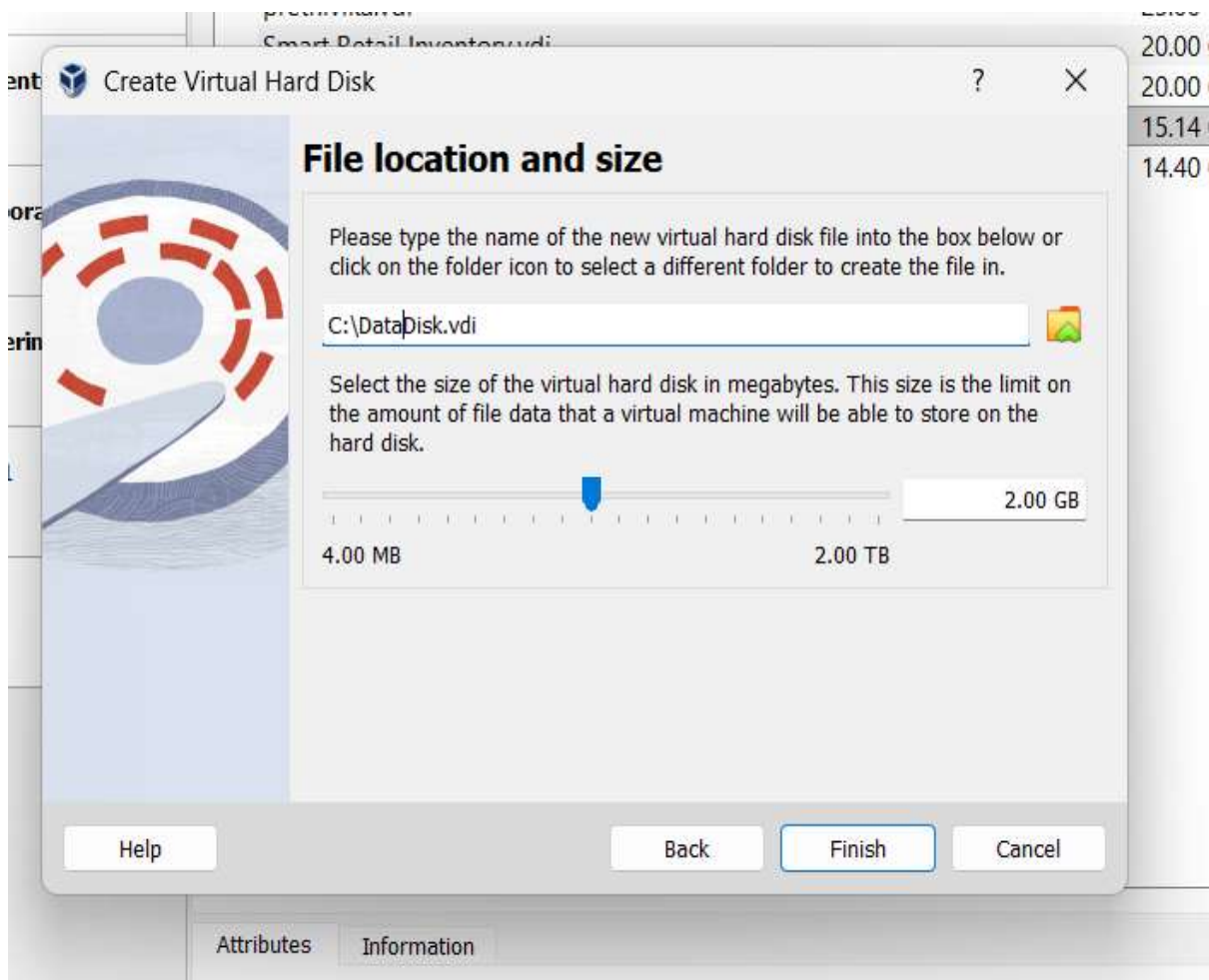
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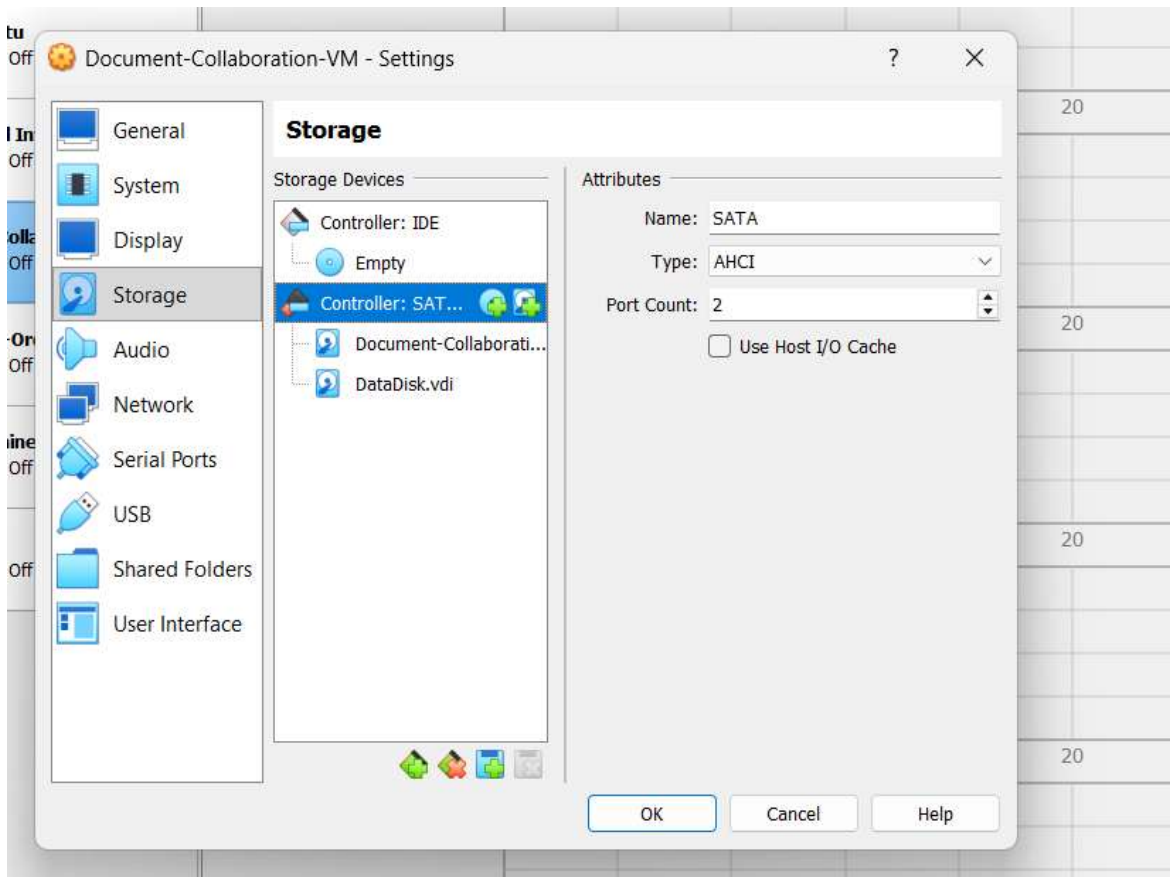
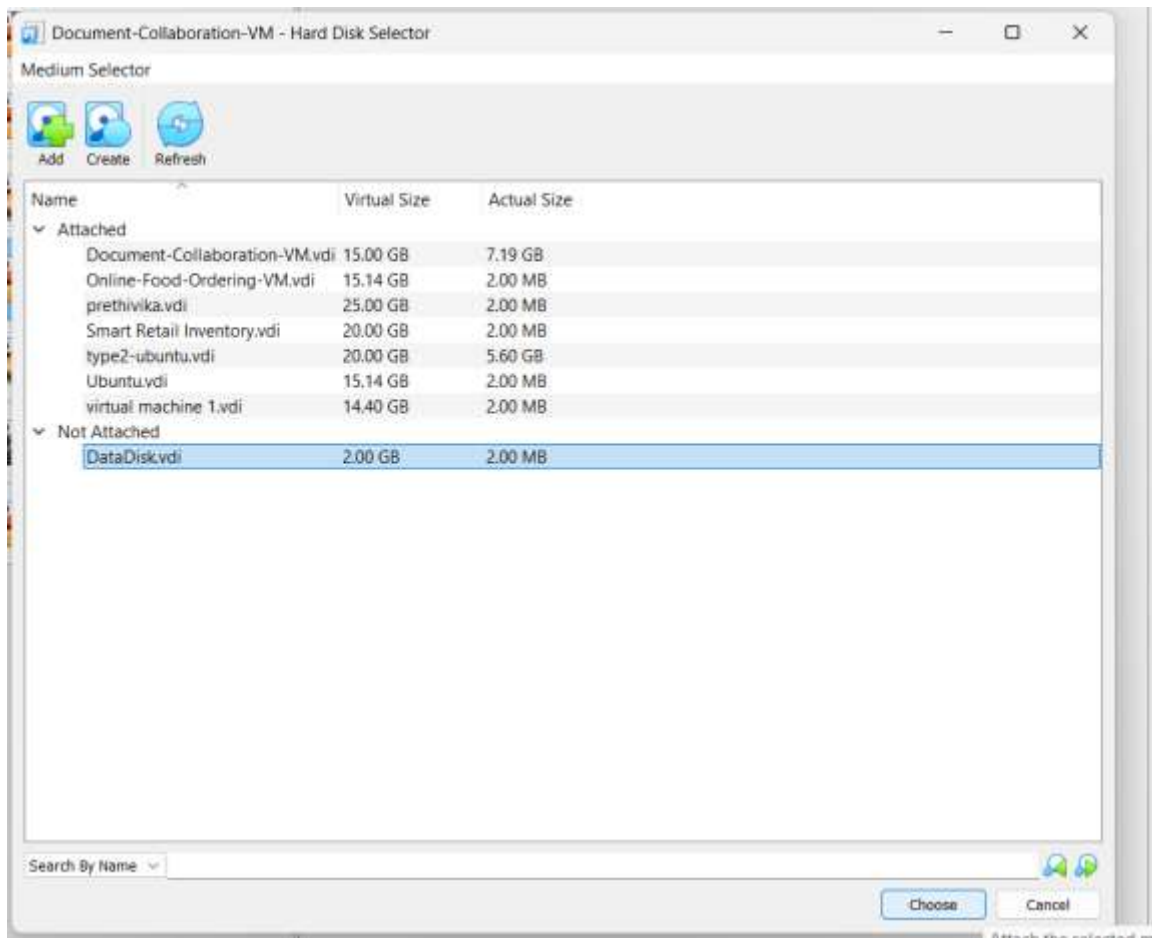
To attach a virtual block storage disk to a virtual machine using **Oracle VirtualBox**, store data on the virtual disk, detach it from the virtual machine, and verify that the data is retained after the disk is attached again.

PROCEDURE

1. Open **Oracle VirtualBox**, select the existing Ubuntu virtual machine, and ensure that the VM is completely powered off.
2. Create a new virtual hard disk using **Tools → Media → Create**, select **VDI (VirtualBox Disk Image)**, choose **Dynamically Allocated**, and assign approximately **5 GB** of storage.
3. Select the Ubuntu VM and open **Settings → Storage**. Add the newly created virtual disk to the **SATA Controller** and save the configuration.
4. Start the Ubuntu VM, open the Terminal, identify the newly attached disk using `lsblk`, format it with the **EXT4** file system, and mount it to a suitable directory.
5. Create a test file on the mounted virtual disk and verify its contents using terminal commands. This confirms that data has been successfully stored on the virtual block device.
6. Shut down the Ubuntu VM and go to **Settings → Storage** in VirtualBox. Select the virtual disk and choose **Remove Attachment** without deleting the virtual disk file.
7. Attach the **same virtual disk** to the Ubuntu VM again, start the VM, mount the disk, and verify that the previously created test file and its data are still available.

IMPLEMENTATION:





RESULT

The virtual block storage disk was successfully attached to the Ubuntu virtual machine, and data was stored on it. After detaching and reattaching the virtual disk, the stored data was successfully recovered, demonstrating that the virtual block storage retains data independently of its attachment to the virtual machine.